Exact quantum–classical decoupling on the Hilbert slice

A refinement adjacent to arXiv:2407.17109

Autonomous proof packet — agent_lane_18 (GPT5.6)

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Abstract

For every finite frequency partition (indeed, every finite family of frequency sets) and every $1 \le q \le \infty$, the classical and quantum decoupling constants at output exponent $p = 2$ coincide exactly:

$$\mathcal{D}^{\mathscr{Q}}_{2,q}(\mathcal{P}_\Omega) = \mathcal{D}^{\mathscr{C}}_{2,q}(\mathcal{P}_\Omega).$$

The source paper records only the case $p = q = 2$. The proof is an isometric identification: $U = \mathcal{F}_\sigma \mathcal{F}_W$ is a unitary map from Hilbert–Schmidt operators onto L^2, and it preserves the localization constraint term by term. If $\mathcal{P}_\Omega$ has N pairwise disjoint positive-measure pieces, the common optimal constant is explicitly

$$1 \quad (1 \le q \le 2), \qquad N^{1/2-1/q} \quad (2 \le q \le \infty).$$

This is a candidate new exact refinement, not a solution of the queue's literal "natural question," which the paper itself immediately answers.

1 Source statement and scope

Samuelsen, arXiv:2407.17109v2, asks on page 2:

> A natural question is therefore if similar results can be achieved for decoupling.

The next sentence begins "To answer this question," and Theorem 1.3 proves equivalence of the classical and quantum constants up to a factor depending only on the bounded ambient set. Thus the queue signal is not an open problem. The same page then makes the more precise optimality observation shown in the source crop below.

> Rather interesting is the fact that the constant in Theorem 1.3 relies only on Ω and not on the partition $\mathcal{P}_\Omega$. The fact that Ω is bounded is crucial. The proof uses a quantum version of Wiener's division lemma. More specifically, we use the fact that any function whose Fourier transform is compactly supported can be written as a convolution between itself and a suitable Schwartz function.
>
> Since the constant $C(\Omega)$ is the same for all choices of p and q, it is unlikely to be optimal. For instance, whenever $p = q = 2$, the classical and quantum decoupling constants coincide by Plancherel's Theorem. Hence, we must have $C(\Omega) \equiv 1$ for the constant to be optimal in this case.

The relevant sentence is:

> For instance, whenever $p = q = 2$, the classical and quantum decoupling constants coincide by Plancherel's Theorem.

The result below strengthens this from $q = 2$ to every q and computes the constant for a genuine finite partition. No boundedness assumption on Ω is needed for this Hilbert-space statement.

2 Definitions and the isometric correspondence

Let $\mathcal{P} = \{\theta_1, \ldots, \theta_N\}$ be a finite family of measurable subsets of $\mathbb{R}^{2d}$. The classical constant $\mathcal{D}^{\mathscr{C}}_{2,q}(\mathcal{P})$ is the least D such that

$$\left\| \sum_{j=1}^{N} f_j \right\|_2 \leq D \left\| (\|f_j\|_2)_{j=1}^N \right\|_{\ell^q}$$

whenever $f_j \in L^2(\mathbb{R}^{2d})$ and $\operatorname{supp}(\mathcal{F}_\sigma f_j) \subseteq \theta_j$. The quantum constant $\mathcal{D}^{\mathscr{Q}}_{2,q}(\mathcal{P})$ is defined by the analogous inequality for Hilbert–Schmidt operators T_j, with $\|\cdot\|_2$ replaced by the Hilbert–Schmidt norm and localization $\operatorname{supp}(\mathcal{F}_W T_j) \subseteq \theta_j$. These are precisely the finite-norm objects in the source definitions when $p = 2$.

The source normalizations give two unitary maps:

$$\mathcal{F}_W : \mathcal{S}^2 \longrightarrow L^2(\mathbb{R}^{2d}), \qquad \mathcal{F}_\sigma : L^2(\mathbb{R}^{2d}) \longrightarrow L^2(\mathbb{R}^{2d}), \qquad \mathcal{F}_\sigma^2 = I.$$

Consequently

$$U := \mathcal{F}_\sigma \mathcal{F}_W : \mathcal{S}^2 \longrightarrow L^2(\mathbb{R}^{2d})$$

is unitary, and for every $T \in \mathcal{S}^2$,

$$\mathcal{F}_\sigma(UT) = \mathcal{F}_\sigma^2(\mathcal{F}_W T) = \mathcal{F}_W T. \tag{1}$$

Thus U preserves the localization set exactly, rather than merely enlarging it.

3 Equality for every inner exponent

Theorem 1 (exact Hilbert-slice transference)**.** *For every finite family $\mathcal{P}$ of measurable frequency sets and every $1 \leq q \leq \infty$,*

$$\boxed{\mathcal{D}^{\mathscr{Q}}_{2,q}(\mathcal{P}) = \mathcal{D}^{\mathscr{C}}_{2,q}(\mathcal{P}).}$$

The statement remains true when the sets overlap.

Proof. Take any admissible quantum collection $(T_j)_{j=1}^N$ and set $f_j = UT_j$. Equation (1) makes (f_j) classically admissible, while unitarity and linearity give

$$\left\| \sum_j f_j \right\|_2 = \left\| \sum_j T_j \right\|_{\mathcal{S}^2}, \qquad \|f_j\|_2 = \|T_j\|_{\mathcal{S}^2} \quad (1 \leq j \leq N).$$

Applying the classical inequality therefore proves $\mathcal{D}^{\mathscr{Q}}_{2,q} \leq \mathcal{D}^{\mathscr{C}}_{2,q}$, including the usual maximum convention for $q = \infty$.

Conversely, U is onto and

$$U^{-1} f = \mathcal{F}_W^{-1}(\mathcal{F}_\sigma f).$$

For every classically admissible (f_j), set $T_j = U^{-1} f_j$. Then $\mathcal{F}_W T_j = \mathcal{F}_\sigma f_j$, so the same identities and the quantum inequality yield $\mathcal{D}^{\mathscr{C}}_{2,q} \leq \mathcal{D}^{\mathscr{Q}}_{2,q}$. The two optimal constants are equal. □

4 Exact value for a disjoint partition

Theorem 2 (sharp finite-partition formula)**.** *Suppose $\theta_1, \ldots, \theta_N$ are pairwise disjoint up to null sets and each has positive Lebesgue measure. Then*

$$\mathcal{D}^{\mathscr{C}}_{2,q}(\mathcal{P}) = \mathcal{D}^{\mathscr{Q}}_{2,q}(\mathcal{P}) = \begin{cases} 1, & 1 \leq q \leq 2, \\ N^{\frac{1}{2} - \frac{1}{q}}, & 2 \leq q \leq \infty, \end{cases}$$

where $1/\infty = 0$.

Proof. If (f_j) is admissible, then the functions $\mathcal{F}_\sigma f_j$ have pairwise disjoint supports. Plancherel therefore makes the f_j pairwise orthogonal, and with $a_j = \|f_j\|_2$,

$$\left\|\sum_j f_j\right\|_2 = \left(\sum_j a_j^2\right)^{1/2} = \|a\|_{\ell^2}.$$

The optimal upper bound is exactly the norm of the identity map $\ell_N^q \to \ell_N^2$. Monotonicity of finite-dimensional ℓ^r norms gives norm 1 for $q \le 2$; Hölder gives $N^{1/2-1/q}$ for $q \ge 2$.

These constants are sharp within the localized class. For $q \le 2$, choose one nonzero component. For $q \ge 2$, positive measure lets us choose $0 \ne g_j \in L^2$ supported in θ_j, normalized by $\|g_j\|_2 = 1$, and put $f_j = \mathcal{F}_\sigma g_j$. Then every $a_j = 1$, so equality holds in the stated finite-dimensional norm comparison. The quantum sharpness follows either from Theorem 1 or by applying U^{-1} term by term. ☐

Corollary 1 (infinitely many pieces)**.** *For a countable pairwise-disjoint family with infinitely many positive-measure pieces, uniform decoupling over its finite subfamilies has constant* 1 *for* $1 \le q \le 2$ *and no finite constant for* $q > 2$*.*

Proof. Theorem 2 applied to arbitrary N-element subfamilies gives the claim, since $N^{1/2-1/q} \to \infty$ exactly when $q > 2$. ☐

5 Verification, novelty boundary, and limitations

The proof uses only three identities stated in the latest source version: $\mathcal{F}_W : \mathcal{S}^2 \to L^2$ is unitary, $\mathcal{F}_\sigma$ is unitary, and $\mathcal{F}_\sigma^2 = I$. Direct substitution gives (1); hence there is no convolution loss, support enlargement, density passage, or dependence on Ω. The endpoint checks are also explicit: at $q = 1$ the constant is 1, at $q = 2$ it is 1, and at $q = \infty$ it is $\sqrt{N}$.

A search of the run's lightweight result indexes and targeted searches for the exact all-q identity found the source's $p = q = 2$ remark but no statement of Theorems 1–2. This is evidence, not a literature guarantee. Human review should verify novelty against the published version and standard quantum-harmonic-analysis references.

This packet does not improve the comparison for $p \ne 2$. It does not answer a genuinely open question quoted from the paper: the queue's "natural question" is answered by Theorem 1.3 of the same paper. The promoted result is instead an exact, fully proved strengthening of the adjacent optimality remark.

References

1. H. J. Samuelsen, *Decoupling for Schatten class operators in the setting of Quantum Harmonic Analysis*, arXiv:2407.17109v2 (2024), published in the Bulletin of the London Mathematical Society.

2. J. C. T. Pool, *Mathematical aspects of the Weyl correspondence*, Journal of Mathematical Physics 7 (1966), cited by the source for the unitarity of the Fourier–Wigner transform on Hilbert–Schmidt operators.